Tangent-Process Regression Targets for Martingale Extensions and Logarithmic Transforms of Diffusions

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Abstract

Given a diffusion $dX_s = f(X_s, s)\,ds + \sigma(X_s, s)\,dW_s$ on $[0, 1]$ and a terminal function h, we derive simulation-based regression targets for the martingale extension $h(x, t) = \mathbb{E}[h(X_1) \mid X_t = x]$ *and its spatial gradient*, using the first-variation (tangent) process of the flow. A single trajectory, together with one reverse-mode sweep through the unrolled simulator, produces unbiased targets for both the value and the full n-dimensional gradient at every time step, in contrast to standard Euler schemes for the associated BSDE which use only scalar value information. We then extend the construction to the logarithmic transform $v(x, t) = \log \mathbb{E}[e^{r(X_1)} \mid X_t = x]$, whose BSDE has a quadratic driver. The key structural facts are: (i) under the Doob h-transform the driver vanishes and the gradient admits a driver-free tangent-process representation along tilted paths; (ii) the tilted dynamics can be realized by a shifted-noise rollout with a *detached* control, so that reverse-mode differentiation again yields the exact discrete targets; (iii) the importance weights correcting for an approximate control are *pathwise* (not merely in expectation) equal to one at the fixed point; and (iv) the value targets are biased from below by exactly half the squared L^2 control error, which is the Boué–Dupuis variational principle. Proofs and references are given.

Contents

1 Setting and assumptions

Let $(\Omega, \mathcal{F}, (\mathcal{F}_s)_{s\in[0,1]}, \mathbb{P})$ carry an m-dimensional Brownian motion W, and consider the Itô SDE in $\mathbb{R}^n$

$$dX_s = f(X_s, s)\, ds + \sigma(X_s, s)\, dW_s, \qquad s \in [t, 1], \qquad X_t = x, \tag{1}$$

with solution denoted $X_s^{t,x}$. We write $\sigma_k(\cdot, s) \in \mathbb{R}^n$ for the k-th column of $\sigma \in \mathbb{R}^{n\times m}$, $a = \sigma\sigma^\top$, and $\nabla f \in \mathbb{R}^{n\times n}$ for the spatial Jacobian, $(\nabla f)_{jk} = \partial_{x_k} f_j$.

Assumption 1.1. f and σ are continuous in (x, s), continuously differentiable in x with bounded spatial derivatives, and of linear growth. The terminal datum (h in Section 2, r in Section 3) is C^1 with $h, \nabla h$ (resp. $r, \nabla r$) of polynomial growth; in Section 3 we additionally assume r bounded above (so that $\mathbb{E}[e^{r(X_1)}] < \infty$; boundedness of r suffices for the quadratic BSDE theory we cite, and [6] relaxes this to exponential moments).

Under Assumption 1.1 the map $x \mapsto X_s^{t,x}$ is, for a.e. ω, a C^1 diffeomorphism onto its image (Kunita [1, Thm. 4.6.5]), and the Jacobian *tangent process* $J_{t\to s} := \partial X_s^{t,x}/\partial x \in \mathbb{R}^{n\times n}$ satisfies the variational SDE

$$dJ_{t\to s} = \nabla f(X_s, s)\, J_{t\to s}\, ds + \sum_{k=1}^{m} \nabla\sigma_k(X_s, s)\, J_{t\to s}\, dW_s^k, \qquad J_{t\to t} = I_n. \tag{2}$$

By Gronwall's lemma together with the Burkholder–Davis–Gundy inequality, boundedness of $\nabla f, \nabla\sigma_k$ yields

$$\sup_x \mathbb{E}\Big[\sup_{s\in[t,1]} \|J_{t\to s}\|^p\Big] < \infty \qquad \text{for every } p \geq 1, \tag{3}$$

with a constant depending only on p and the Lipschitz bounds (see [1, Sec. 4.6]).

2 The zero-driver case: martingale extension of h

Define

$$h(x, t) := \mathbb{E}\big[h(X_1^{t,x})\big], \tag{4}$$

so that $h(X_s, s)$ is a martingale and $h(\cdot, t)$ solves the backward Kolmogorov equation $\partial_t h + f \cdot \nabla h + \frac{1}{2}\operatorname{tr}(a\nabla^2 h) = 0$ with $h(\cdot, 1) = h$.

2.1 Pathwise gradient representation

Theorem 2.1 (Pathwise differentiation of the martingale extension)**.** *Under Assumption 1.1, $h(\cdot, t) \in C^1(\mathbb{R}^n)$ and*

$$\nabla_x h(x, t) \;=\; \mathbb{E}\big[\, J_{t\to 1}^\top\, \nabla h(X_1^{t,x})\,\big]. \tag{5}$$

Proof. Fix t, x and a direction $e \in \mathbb{R}^n$, $\|e\| = 1$. For $\varepsilon > 0$, the mean value theorem applied ω-wise along the C^1 flow gives

$$\frac{h(X_1^{t,x+\varepsilon e}) - h(X_1^{t,x})}{\varepsilon} = \int_0^1 \nabla h\big(X_1^{t,x+\theta\varepsilon e}\big)^\top\, \partial_x X_1^{t,x+\theta\varepsilon e}\, e\, d\theta.$$

As $\varepsilon \downarrow 0$ the integrand converges a.s. to $\nabla h(X_1^{t,x})^\top J_{t\to 1}\, e$ by continuity of the flow and of its derivative in the initial condition [1, Thm. 4.6.5]. Domination: by polynomial growth of ∇h, the standard moment bound $\sup_{|y-x|\le 1}\mathbb{E}[\sup_s |X_s^{t,y}|^p] < \infty$, and (3), the family of integrands is bounded in $L^q(\mathbb{P}\otimes d\theta)$ for some $q > 1$ (Cauchy–Schwarz), hence uniformly integrable. Vitali's convergence theorem lets us pass to the limit inside the expectation, giving the directional derivative $\partial_e h(x,t) = \mathbb{E}[\nabla h(X_1)^\top J_{t\to 1} e]$. Continuity of the right-hand side in x (same uniform integrability plus a.s. continuity of $x \mapsto (X_1^{t,x}, J_{t\to 1})$) upgrades directional differentiability to C^1, and (5) follows by transposition. For the BSDE phrasing of this representation theorem, including non-Markovian terminal data, see Ma and Zhang [4]. □

2.2 BSDE and Malliavin interpretation

Setting $Y_s = h(X_s, s)$ and $Z_s = \sigma(X_s, s)^\top \nabla h(X_s, s)$, Itô's formula and the Kolmogorov equation give the (zero-driver) BSDE

$$dY_s = Z_s^\top dW_s, \qquad Y_1 = h(X_1), \tag{6}$$

the linear special case of Pardoux–Peng [2]; see also [3]. The classical representation $Z_t = \mathbb{E}[D_t Y_1 \mid \mathcal{F}_t]$ (Clark–Ocone / [3, 4]), combined with the chain rule $D_t h(X_1) = \nabla h(X_1)^\top D_t X_1$ and the flow identity $D_t X_1 = J_{t\to 1}\,\sigma(X_t, t)$, recovers (5) in the form $Z_t = \sigma^\top \nabla h(X_t, t)$. The point of view that the pair (Y, Z) jointly satisfies a vector BSDE with terminal condition $(h(X_1), \sigma^\top \nabla h(X_1))$ — so that the *known* terminal gradient is data, not a by-product — is developed as the One-Step Malliavin (OSM) scheme by Negyesi, Andersson and Oosterlee [14].

2.3 Discrete targets by reverse-mode differentiation

Fix a grid $t = t_0 < t_1 < \cdots < t_N = 1$ and the Euler–Maruyama recursion

$$\widehat{X}_{i+1} = \widehat{X}_i + f(\widehat{X}_i, t_i)\,\Delta t_i + \sigma(\widehat{X}_i, t_i)\,\Delta W_i, \qquad \Delta W_i \sim \mathcal{N}(0, \Delta t_i I_m) \text{ i.i.d.} \tag{7}$$

Freezing the increments $(\Delta W_i)_i$, the map $\widehat{X}_i \mapsto \widehat{X}_N$ is a composition of smooth maps with step Jacobians

$$\widehat{J}_{i+1\leftarrow i} = I + \nabla f(\widehat{X}_i, t_i)\,\Delta t_i + \sum_k \nabla \sigma_k(\widehat{X}_i, t_i)\,\Delta W_i^k, \tag{8}$$

which is exactly the Euler discretization of (2).

Proposition 2.2 (Backprop computes the discrete tangent targets)**.** *Let $g_i \in \mathbb{R}^n$ denote the gradient delivered to the intermediate node $\widehat{X}_i$ by reverse-mode automatic differentiation of the scalar $h(\widehat{X}_N)$ through the unrolled recursion (7) with (ΔW_i) held fixed. Then*

$$g_i = \widehat{J}_{N\leftarrow i}^\top \nabla h(\widehat{X}_N), \qquad \widehat{J}_{N\leftarrow i} := \widehat{J}_{N\leftarrow N-1} \cdots \widehat{J}_{i+1\leftarrow i},$$

i.e. the exact gradient of the discrete *value function $\widehat{h}_i(x) := \mathbb{E}[h(\widehat{X}_N) \mid \widehat{X}_i = x]$ is $\nabla \widehat{h}_i(x) = \mathbb{E}[g_i \mid \widehat{X}_i = x]$. Moreover one reverse sweep computes $(g_0, \ldots, g_{N-1})$ simultaneously, at a cost of $O(1)$ rollouts.*

Proof. The first identity is the chain rule: reverse-mode AD computes vector–Jacobian products, and the Jacobian of the frozen-noise composition $\widehat{X}_i \mapsto \widehat{X}_N$ is $\widehat{J}_{N\leftarrow i}$ by (8) and the product rule. The second identity is Theorem 2.1 applied verbatim to the discrete-time Markov chain: differentiation under the expectation is justified by the same uniform-integrability argument, the moment bound (3) holding for products of the matrices (8) with constants uniform in N (discrete Gronwall). The cost claim is the cheap-gradient principle of reverse-mode AD [12]: the backward sweep visits each node once and deposits its adjoint there. □

Remark 2.3 (Consistency). The targets g_i are the *exact* gradients of the discrete flow that the scheme itself simulates, not $O(\Delta t)$ approximations of the continuous tangent. The regression therefore learns the gradient of the same discrete value function whose values it learns; discretization error enters only through $\widehat{h} \approx h$, with the usual weak-order-1 rate.

2.4 Sobolev regression

Since, conditionally on $\widehat{X}_i$, the pair $\big(h(\widehat{X}_N),\, g_i\big)$ is unbiased for $\big(\widehat{h}(\widehat{X}_i, t_i),\, \nabla\widehat{h}(\widehat{X}_i, t_i)\big)$, the population loss

$$\mathcal{L}(\theta) = \mathbb{E} \sum_{i=0}^{N-1} \Big[\big(h_\theta(\widehat{X}_i, t_i) - h(\widehat{X}_N)\big)^2 + \lambda \, \|\nabla_x h_\theta(\widehat{X}_i, t_i) - g_i\|^2 \Big] \tag{9}$$

is minimized, for every $\lambda > 0$, by $h_\theta = \widehat{h}$ (each term is a conditional L^2 projection with the same minimizer; λ trades off variance and conditioning of the optimization, not bias). Each trajectory supplies $(n+1)N$ scalar targets at the cost of one rollout and one reverse sweep. This is Sobolev training; in the financial setting with regression at t_0 only it is the “differential machine learning” of Huge and Savine [13], and the all-times version coincides with the driver-free specialization of [14].

Remark 2.4 (Variance; nonsmooth h). The targets $J^\top \nabla h(X_1)$ may be heavy-tailed when the flow is expansive ($\|\nabla f\|$ large, multiplicative noise); antithetic increments and the iterated control variate of [15] (subtract $\sum_i \nabla h_\theta(\widehat{X}_i)^\top \sigma \, \Delta W_i$ from the value target, which by (6) is exact at convergence) mitigate this. If ∇h is unavailable, the Bismut–Elworthy–Li formula [11] $\nabla h(x,t) = \mathbb{E}\big[h(X_1)\, \frac{1}{1-t} \int_t^1 (\sigma^{-1}(X_s, s) J_{t\to s})^\top dW_s\big]$ (for elliptic σ) trades the terminal gradient for an integration-by-parts weight, at the cost of variance of order $(1-t)^{-1}$.

3 The logarithmic transform: quadratic driver

Now let

$$v(x,t) := \log \mathbb{E}\big[e^{r(X_1^{t,x})}\big], \qquad u := e^v, \tag{10}$$

so $u(X_s, s)$ is a positive martingale, u solves the Kolmogorov equation, and the logarithmic (Hopf–Cole / Fleming) transform [9, Ch. VI] yields the Hamilton–Jacobi–Bellman equation

$$\partial_t v + f \cdot \nabla v + \tfrac{1}{2} \operatorname{tr}\big(a \, \nabla^2 v\big) + \tfrac{1}{2} \, \|\sigma^\top \nabla v\|^2 = 0, \qquad v(\cdot, 1) = r. \tag{11}$$

With $Y_s = v(X_s, s)$, $Z_s = \sigma(X_s, s)^\top \nabla v(X_s, s)$, Itô’s formula gives the quadratic BSDE

$$dY_s = -\tfrac{1}{2} \|Z_s\|^2 \, ds + Z_s^\top dW_s, \qquad Y_1 = r(X_1), \tag{12}$$

well-posed for bounded r with $Z \in \mathrm{BMO}$ by Kobylanski [5] (unbounded r with exponential moments: [6]).

Applying Section 2 directly to $u = e^v$ is correct ($\nabla u(x,t) = \mathbb{E}[e^{r(X_1)} J_{t\to 1}^\top \nabla r(X_1)]$) but suffers exponential-in-r relative variance, the log-partition-function pathology. The remedy is to absorb the exponential weight into the dynamics.

3.1 The Doob h-transform removes the driver

Since $u(X_s, s)$ is a positive martingale, Itô on $\log u$ gives the Doléans exponential representation

$$\frac{u(X_s, s)}{u(x,t)} = \exp\Big(\int_t^s Z_\rho^\top dW_\rho - \tfrac{1}{2}\int_t^s \|Z_\rho\|^2 d\rho\Big), \qquad s \in [t,1]. \tag{13}$$

Define, conditionally on $X_t = x$, the tilted measure

$$\frac{d\mathbb{Q}}{d\mathbb{P}}\Big|_{\mathcal{F}_1} = e^{\,r(X_1)-v(x,t)} = \frac{u(X_1,1)}{u(x,t)}. \tag{14}$$

By (13) and Girsanov's theorem ($Z \in$ BMO guarantees the Doléans exponential is a true martingale [7]), $W_s^{\mathbb{Q}} := W_s - \int_t^s Z_\rho \, d\rho$ is a $\mathbb{Q}$-Brownian motion and the h-transformed dynamics are

$$dX_s = \big(f + a\,\nabla v\big)(X_s, s)\,ds + \sigma(X_s, s)\,dW_s^{\mathbb{Q}}. \tag{15}$$

This is the classical equivalence between the logarithmic transform and entropy-regularized stochastic control [9, 10].

Theorem 3.1 (Driver-free gradient representation under the tilt)**.** *Under Assumption 1.1,*

$$\nabla v(x,t) \;=\; \mathbb{E}_{\mathbb{Q}}\big[\, J_{t\to1}^\top \nabla r(X_1) \,\big|\, X_t = x\big], \tag{16}$$

where $J_{t\to s}$ is the tangent process (2) of the uncontrolled *flow, i.e. the solution of $dJ = \nabla f\, J\, ds + \sum_k \nabla\sigma_k\, J\, dW_s^k$ driven by the original increments $dW_s = dW_s^{\mathbb{Q}} + Z_s\,ds$, evaluated along the tilted path (15).*

Proof 1 (change of measure). Apply Theorem 2.1 to the terminal function e^r (polynomial growth of $e^r\,\nabla r$ holds since r is bounded above and ∇r has polynomial growth): $\nabla u(x,t) = \mathbb{E}_{\mathbb{P}}\big[e^{r(X_1)}\, J_{t\to1}^\top \nabla r(X_1)\big]$. Divide by $u(x,t)$ and use $\nabla v = \nabla u/u$ together with (14):

$$\nabla v(x,t) = \mathbb{E}_{\mathbb{P}}\Big[e^{\,r(X_1)-v(x,t)}\, J_{t\to1}^\top \nabla r(X_1)\Big] = \mathbb{E}_{\mathbb{Q}}\big[J_{t\to1}^\top \nabla r(X_1)\big].$$

The pair (X, J) is a path functional of (x, W); rewriting its driving noise as $dW = dW^{\mathbb{Q}} + Z\,dt$ expresses the same functional in terms of the $\mathbb{Q}$-Brownian motion, which is the stated form. ☐

Proof 2 (PDE / martingale argument; σ constant for clarity). Let $p := \nabla v$. Differentiating (11) in x_k and using symmetry of the Hessian, $\partial_k p_j = \partial_j p_k$, so that $p^\top a\,\partial_k p = (ap)\cdot\nabla p_k$:

$$\partial_t p_k + \big(f + ap\big)\cdot\nabla p_k + \tfrac{1}{2}\operatorname{tr}\big(a\nabla^2 p_k\big) = -\big((\nabla f)^\top p\big)_k. \tag{17}$$

The quadratic term has recombined with f into the *controlled* drift of (15): in BSDE language, a driver whose z-gradient is $\nabla_z(\frac{1}{2}\|z\|^2) = z$ is absorbed exactly by the Girsanov shift with kernel Z, which is the h-transform. Now set $M_s := J_{t\to s}^\top\, p(X_s, s)$ along the tilted dynamics. With σ constant, $dJ = \nabla f(X_s, s)\,J\,ds$ under either measure, and Itô's formula under $\mathbb{Q}$ together with (17) gives

$$dM_s = J^\top(\nabla f)^\top p\,ds + J^\top\Big(\underbrace{\partial_t p + (f+ap)\cdot\nabla p + \tfrac{1}{2}\operatorname{tr}(a\nabla^2 p)}_{=-(\nabla f)^\top p}\Big)\,ds + J^\top(\nabla p)^\top\sigma\,dW_s^{\mathbb{Q}} = J^\top(\nabla p)^\top\sigma\,dW_s^{\mathbb{Q}},$$

a local martingale; the moment bounds (3) (valid under $\mathbb{Q}$ since $f + a\nabla v$ inherits local Lipschitz bounds on compacts, with a standard localization) make it a true martingale on $[t,1]$, whence $p(x,t) = M_t = \mathbb{E}_{\mathbb{Q}}[M_1] = \mathbb{E}_{\mathbb{Q}}[J_{t\to1}^\top \nabla r(X_1)]$. For state-dependent σ, Proof 1 applies unchanged. ☐

Remark 3.2. Equation (16) says: *under the tilted measure, the gradient process obeys the same driver-free tangent representation as in Section 2.* All the machinery of Sections 2.3–2.4 therefore transfers, provided we can sample (approximately) from $\mathbb{Q}$ and correct for the approximation. This is the BSDE-side derivation of *Adjoint Matching* [18]; the importance-weighted version with general gauge freedom is *Stochastic Optimal Control Matching* [17], of which the present problem is the terminal-cost-only special case.

3.2 Discrete realization: shifted noise, detached control

Let v_θ be the current model and $\hat{u}_\theta := \sigma^\top \nabla v_\theta$ the induced control. Simulate the controlled chain by *shifting the noise*:

$$\widehat{X}_{i+1} = \widehat{X}_i + f(\widehat{X}_i, t_i)\,\Delta t_i + \sigma(\widehat{X}_i, t_i)\,\Delta\widetilde{W}_i, \qquad \Delta\widetilde{W}_i := \Delta W_i + \hat{u}_\theta(\widehat{X}_i, t_i)\,\Delta t_i, \tag{18}$$

with $\Delta W_i \sim \mathcal{N}(0, \Delta t_i I)$ the increments of the sampling (controlled-measure) Brownian motion and $\hat{u}_\theta$ treated as an *exogenous constant* in the computation graph ("detached").

Proposition 3.3 (Detached backprop computes the uncontrolled tangent along tilted paths)**.** *Reverse-mode differentiation of $r(\widehat{X}_N)$ through (18), with (ΔW_i) and the values $\hat{u}_\theta(\widehat{X}_i, t_i)$ held fixed, delivers to node $\widehat{X}_i$ the vector*

$$g_i = \widehat{J}_{N\leftarrow i}^\top \nabla r(\widehat{X}_N), \qquad \widehat{J}_{i+1\leftarrow i} = I + \nabla f\,\Delta t_i + \sum_k \nabla\sigma_k\,\Delta\widetilde{W}_i^k,$$

i.e. the Euler discretization of the uncontrolled *variational SDE (2) driven by the shifted increments $\Delta\widetilde{W}_i$ — precisely the discretization of the process appearing in Theorem 3.1, since $dW = dW^{\mathbb{Q}} + Z\,dt \approx dW^{\hat{u}} + \hat{u}\,dt$. In particular, when $\hat{u}_\theta = \sigma^\top \nabla v$, the g_i are conditionally unbiased (in the discrete model) for $\nabla v(\widehat{X}_i, t_i)$; no second derivatives of v_θ are ever formed.*

Proof. Immediate from the chain rule applied to (18) with $\hat{u}_\theta$ exogenous: the step Jacobian is as displayed, including the cross terms $\nabla\sigma_k\,\hat{u}^k \Delta t$ generated by state-dependent σ multiplying the shifted increment. Detaching is not an approximation but the content of Theorem 3.1: the correct integrand is the tangent of the *original* flow along the tilted path. Differentiating through the control would add terms proportional to $\nabla\hat{u}_\theta = \nabla(\sigma^\top \nabla v_\theta)$, i.e. spurious Hessians, and the resulting expectation would no longer equal (16). Equivalently, the backward sweep solves the *lean adjoint* equation $d\tilde{a}_s = -\nabla f(X_s, s)^\top \tilde{a}_s\,ds - \sum_k \nabla\sigma_k(X_s, s)^\top \tilde{a}_s\,d\widetilde{W}_s^k$ (backward, $\tilde{a}_1 = \nabla r(X_1)$) of [18]. □

3.3 Importance weights and their pathwise collapse

Since we sample under $\hat{u} = \hat{u}_\theta$ rather than the exact kernel Z, expectations under $\mathbb{Q}$ must be importance-corrected. Let $\mathbb{P}^{\hat{u}}$ denote the law of (15) with $a\nabla v$ replaced by $\sigma\hat{u}$, and $W^{\hat{u}}$ its Brownian motion (the sampled increments). Conditionally on $X_t = x$, combining (14) with the Girsanov factor between $\mathbb{P}$ and $\mathbb{P}^{\hat{u}}$ gives

$$\log w_t \;=\; r(X_1) - v(x,t) - \int_t^1 \hat{u}_s^\top dW_s^{\hat{u}} - \tfrac{1}{2}\int_t^1 \|\hat{u}_s\|^2 ds, \qquad \frac{d\mathbb{Q}}{d\mathbb{P}^{\hat{u}}}\Big|_{\mathcal{F}_1} = w_t. \tag{19}$$

Lemma 3.4 (Pathwise form of the weights)**.** *Along $\mathbb{P}^{\hat{u}}$-paths, with $Z_s = \sigma^\top \nabla v(X_s, s)$ the exact kernel,*

$$\log w_t \;=\; \int_t^1 (Z_s - \hat{u}_s)^\top dW_s^{\hat{u}} \;-\; \tfrac{1}{2}\int_t^1 \|Z_s - \hat{u}_s\|^2\, ds. \tag{20}$$

In particular $\log w_t \equiv 0$ pathwise *(not merely in expectation) iff $\hat{u}_s = Z_s$ for a.e. s along the path.*

Proof. Under $\mathbb{P}^{\hat{u}}$, $dW_s = dW_s^{\hat{u}} + \hat{u}_s\, ds$, so the BSDE (12) reads $dY_s = \big(Z_s^\top \hat{u}_s - \frac{1}{2}\|Z_s\|^2\big)ds + Z_s^\top dW_s^{\hat{u}}$. Integrating from t to 1 with $Y_t = v(x,t)$, $Y_1 = r(X_1)$, and substituting $r(X_1) - v(x,t)$ into (19):

$$\log w_t = \int_t^1 \Big(Z^\top \hat{u} - \tfrac{1}{2}\|Z\|^2 - \tfrac{1}{2}\|\hat{u}\|^2\Big) ds + \int_t^1 (Z-\hat{u})^\top dW^{\hat{u}} = -\tfrac{1}{2}\int_t^1 \|Z-\hat{u}\|^2 ds + \int_t^1 (Z-\hat{u})^\top dW^{\hat{u}}.$$

□

The weights are thus the Doléans exponential of the *residual* control error: as $v_\theta \to v$ they flatten to 1 pathwise and the effective sample size of self-normalized importance sampling tends to N_{paths}. In practice one replaces the unknown $v(x,t)$ in (19) by $v_\theta(x,t)$ — a function of the conditioning point only, which changes neither the conditional weighted-regression minimizer nor the $O(1)$ scale of the weights — self-normalizes within each time slice, and, if the effective sample size degrades early in training, resamples: this is twisted sequential Monte Carlo with twist e^{v_θ}.

3.4 Value targets and the Boué–Dupuis bound

Proposition 3.5 (Exact bias of the controlled value target)**.** *Define, along $\mathbb{P}^{\hat{u}}$-paths,*

$$V_t \coloneqq r(X_1) - \tfrac{1}{2}\int_t^1 \|\hat{u}_s\|^2\, ds.$$

Then

$$\mathbb{E}^{\hat{u}}\big[V_t \,\big|\, X_t\big] - v(X_t,t) = -\tfrac{1}{2}\,\mathbb{E}^{\hat{u}}\Big[\int_t^1 \|Z_s - \hat{u}_s\|^2\, ds \,\Big|\, X_t\Big] \;\leq\; 0, \tag{21}$$

with equality iff $\hat{u} = Z$: the bias is second order in the control error and one-sided. Moreover the control-variate-corrected target $V_t - \int_t^1 \hat{u}_s^\top dW_s^{\hat{u}} = v_\theta(X_t,t) + \log w_t^\theta$ (with v_θ used in (19)) has zero variance *at the fixed point, by Lemma 3.4.*

Proof. As in the proof of Lemma 3.4, $r(X_1) - v(X_t,t) = \int_t^1 (Z^\top \hat{u} - \frac{1}{2}\|Z\|^2) ds + \int_t^1 Z^\top dW^{\hat{u}}$. Take conditional expectations (the stochastic integral is a true martingale for $Z \in$ BMO and admissible $\hat{u}$) and subtract $\frac{1}{2}\mathbb{E}\int \|\hat{u}\|^2$:

$$\mathbb{E}^{\hat{u}}[V_t \mid X_t] - v(X_t,t) = \mathbb{E}^{\hat{u}} \int_t^1 \Big(Z^\top \hat{u} - \tfrac{1}{2}\|Z\|^2 - \tfrac{1}{2}\|\hat{u}\|^2\Big) ds = -\tfrac{1}{2}\,\mathbb{E}^{\hat{u}} \int_t^1 \|Z - \hat{u}\|^2 ds.$$

The displayed pathwise identity for the corrected target is (19) rearranged. □

Proposition 3.5 is the conditional form of the Boué–Dupuis variational formula [8]

$$v(x,t) = \sup_{u \in \mathcal{A}} \; \mathbb{E}^u\Big[r(X_1) - \tfrac{1}{2}\int_t^1 \|u_s\|^2\, ds \,\Big|\, X_t = x\Big],$$

with the gradient head of the model playing the role of the (near-optimal) control; the value bias is the Jensen gap of this evidence lower bound, and the optimizer is the h-transform drift [10, 9].

Corollary 3.6 (Fixed point)**.** *Consider the joint weighted Sobolev regression with targets* $\big(V_i^{\mathrm{cv}},\, g_i\big)$ *from Propositions 3.3 and 3.5, weights* (19) *(self-normalized per slice), and rollouts driven by* $\hat{u}_\theta = \sigma^\top \nabla v_\theta$*. Then* $(v, \nabla v)$ *is a fixed point: at* $v_\theta = v$ *the weights equal* 1 *pathwise, the gradient targets are conditionally unbiased for* ∇v *by Theorem 3.1, and the value targets are pathwise exact by Proposition 3.5. (Uniqueness/convergence of the iteration is the content of the analyses in [18, 17]; the present conditioning on* X_t*, rather than on a control-dependent initial law, removes the initial-distribution obstruction discussed there.)*

3.5 Algorithm

Algorithm 1 Tangent-target regression for $v(x,t) = \log \mathbb{E}[e^{r(X_1)} \mid X_t = x]$

given model v_θ; define $\hat{u}_\theta = \sigma^\top \nabla_x v_\theta$ by automatic differentiation
loop
  sample initial points (x_0) and increments $\Delta W_i \sim \mathcal{N}(0, \Delta t_i I)$
  roll out (18) with $\hat{u}_\theta$ **detached**; retain the computation graph
  $g_i \leftarrow$ reverse-mode gradients of $r(\widehat{X}_N)$ at each node $\widehat{X}_i$ ▷ one VJP sweep
  $\log w_i \leftarrow r(\widehat{X}_N) - v_\theta(\widehat{X}_i, t_i) - \sum_{j \ge i} \big[\hat{u}_j^\top \Delta W_j + \frac{1}{2}\|\hat{u}_j\|^2 \Delta t_j\big]$
  $V_i \leftarrow r(\widehat{X}_N) - \sum_{j \ge i} \frac{1}{2}\|\hat{u}_j\|^2 \Delta t_j \ \big[\, - \sum_{j\ge i} \hat{u}_j^\top \Delta W_j$ as control variate $\big]$
  $\bar{w}_i \leftarrow \mathrm{softmax}_{\mathrm{paths}}(\log w_i)$ per time slice (resample if ESS low)
  $\theta \leftarrow \theta - \eta\, \nabla_\theta \sum_i \bar{w}_i \Big[\big(v_\theta(\widehat{X}_i, t_i) - V_i\big)^2 + \lambda\, \|\nabla_x v_\theta(\widehat{X}_i, t_i) - g_i\|^2\Big]$, targets detached
end loop

Remark 3.7 (Practical notes)*.* (i) Use a single network v_θ and obtain ∇v_θ by autodiff (`create_graph=True`), so value and gradient heads are consistent by construction and the learned gradient is conservative. (ii) The controlled drift $f + a\nabla v$ is more expansive than f near maxima of r, so the tangent targets can have heavier tails than in Section 2; antithetic increments and the gauge (reparameterization) matrices M_t of [17] are the standard variance reductions. (iii) Early in training $\hat{u}_\theta$ is uninformative and the weights carry the burden; a warm start with the unweighted scheme of Section 2 applied to $u = e^v$ (or annealing r) stabilizes the loop.

References

[1] H. Kunita. *Stochastic Flows and Stochastic Differential Equations*. Cambridge Studies in Advanced Mathematics 24, Cambridge University Press, 1990.

[2] E. Pardoux and S. Peng. Backward stochastic differential equations and quasilinear parabolic partial differential equations. In *Stochastic Partial Differential Equations and Their Applications*, Lecture Notes in Control and Information Sciences 176, Springer, 1992.

[3] N. El Karoui, S. Peng, and M.-C. Quenez. Backward stochastic differential equations in finance. *Mathematical Finance*, 7(1):1–71, 1997.

[4] J. Ma and J. Zhang. Representation theorems for backward stochastic differential equations. *Annals of Applied Probability*, 12(4):1390–1418, 2002.

[5] M. Kobylanski. Backward stochastic differential equations and partial differential equations with quadratic growth. *Annals of Probability*, 28(2):558–602, 2000.

[6] P. Briand and Y. Hu. BSDE with quadratic growth and unbounded terminal value. *Probability Theory and Related Fields*, 136:604–618, 2006.

[7] N. Kazamaki. *Continuous Exponential Martingales and BMO.* Lecture Notes in Mathematics 1579, Springer, 1994.

[8] M. Boué and P. Dupuis. A variational representation for certain functionals of Brownian motion. *Annals of Probability*, 26(4):1641–1659, 1998.

[9] W. H. Fleming and H. M. Soner. *Controlled Markov Processes and Viscosity Solutions*, 2nd ed. Springer, 2006. (Logarithmic transformations: Ch. VI.)

[10] P. Dai Pra. A stochastic control approach to reciprocal diffusion processes. *Applied Mathematics and Optimization*, 23:313–329, 1991.

[11] K. D. Elworthy and X.-M. Li. Formulae for the derivatives of heat semigroups. *Journal of Functional Analysis*, 125(1):252–286, 1994.

[12] A. Griewank and A. Walther. *Evaluating Derivatives: Principles and Techniques of Algorithmic Differentiation*, 2nd ed. SIAM, 2008.

[13] B. Huge and A. Savine. Differential machine learning. arXiv:2005.02347, 2020.

[14] B. Negyesi, K. Andersson, and C. W. Oosterlee. The One Step Malliavin scheme: new discretization of BSDEs implemented with deep learning regressions. arXiv:2110.05421; *IMA Journal of Numerical Analysis*, 2024.

[15] M. Sabate-Vidales, D. Šiška, and L. Szpruch. Unbiased deep solvers for linear parametric PDEs. arXiv:1810.05094; *Applied Mathematical Finance*, 2021.

[16] J. Han, A. Jentzen, and W. E. Solving high-dimensional partial differential equations using deep learning. *Proceedings of the National Academy of Sciences*, 115(34):8505–8510, 2018.

[17] C. Domingo-Enrich, J. Han, B. Amos, J. Bruna, and R. T. Q. Chen. Stochastic Optimal Control Matching. arXiv:2312.02027, 2023.

[18] C. Domingo-Enrich, M. Drozdzal, B. Karrer, and R. T. Q. Chen. Adjoint Matching: fine-tuning flow and diffusion generative models with memoryless stochastic optimal control. arXiv:2409.08861, 2024.